CSEP 590tv: Quantum Computing

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Today's Menu

Administrivia

Complex Numbers

Bra's Ket's and All That

Quantum Circuits

Administrivia

Changes: slowing down.

Think: Physics without Calculus
Quantum theory with a minimal of linear algebra

In class problems: hardness on the same order of magnitude as the homework problems.

Problem Set 1: has been posted. Anyone who didn't get my email about the first homework being canceled, please let me know and we will arrange accordingly.

Mailing list: sign up on sheet being passed around.

Office Hours: Ioannis Giotis, 5:30-6:30 Wednesday in 430 CSE

Last Week

Last week we saw that there is a big motivation for understanding
quantum computers. BIG PICTURE: understanding quantum information processing machines is the goal of this class!

We also saw that there were there funny postulates describing quantum systems.

This week we will be slowing down and understanding the basic workings of quantum theory by understanding one qubit and two qubit systems.

Quantum Theory's Language

"Complex linear algebra" is the language of quantum theory

Today we will go through this slowly

1. Complex numbers
2. Complex vectors
3. Bras, Kets, and all that

(in class problem)

4. Qubits
5. Measuring Qubits
6. Evolving Qubits

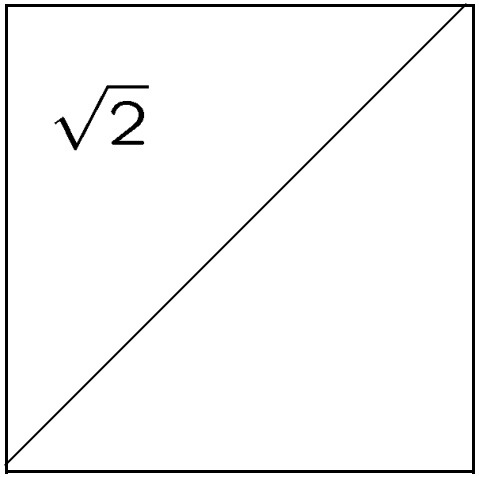
(in class problem)

7. Two qubits: the tensor product
8. Quantum circuits

(in class problem)

Math

Mathematics as a series of discoveries of objects who at first you don't believe exist, and then after you find out they do exist, you discover that they are actually useful!



irrational numbers

Complex Numbers, Definition

Complex numbers are numbers of the form

$$\alpha = a + bi$$

rea rea

"square root of minus one"

$$i = \sqrt{-1} \qquad i^2 = -1$$

Examples: $1 + 4i$

$-1.231 + 23.01i$

$\sqrt{59}$ "purely real"

$\sqrt{-59} = \sqrt{59}i$ "purely imaginary"

roots of $x^2 + x + 1 = 0$

$$x = \frac{-1 \pm \sqrt{1-4}}{2} = \frac{-1}{2} \pm \frac{\sqrt{3}}{2}i$$

Complex Numbers, Geometry

Complex numbers are numbers of the form

$$\alpha = a + bi$$

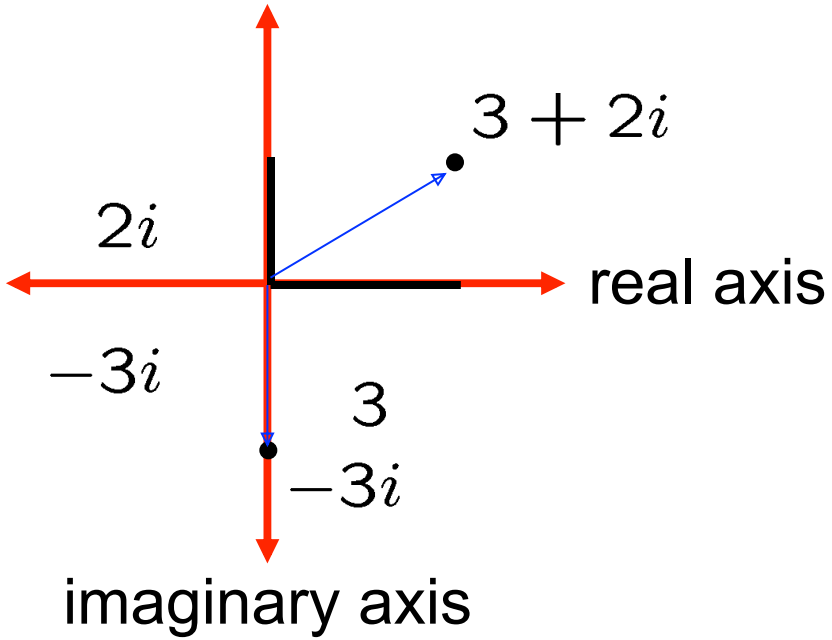
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rea

"square root of minus one"

$$i = \sqrt{-1} \qquad i^2 = -1$$

Complex plane:



Complex Numbers, Math

Complex numbers can be added

$$\alpha = a + bi \qquad \beta = c + di$$

$$(\alpha + \beta) = (a + bi) + (c + di) = (a + b) + (c + d)i$$

Example: $\alpha = 3 + 2i \qquad \beta = 2 - 5i$

$$(\alpha + \beta) = (3 + 2i) + (2 - 5i) = 5 - 3i$$

and multiplied

$$\alpha = a + bi \qquad \beta = c + di$$

$$\alpha\beta = (a + bi)(c + di) = ac + adi + bci + bdi^2$$
$$= (ac - bd) + (ad + bc)i$$

Example: $\alpha = 3 + 2i \qquad \beta = 2 - 5i$

$$\alpha\beta = (3+2i)(2-5i) = 3 \cdot 2 + 3 \cdot (-5i) + 2i \cdot 2 + 2i \cdot (-5i)$$
$$= 16 - 11i$$

Complex Numbers, That * Thing

We can take the complex conjugate of a complex number

$$\alpha = a + bi$$

$$\alpha^* = (a + bi)^* = a - bi$$

Example $\alpha = 3 + 2i$

:

$$\alpha^* = 3 - 2i$$

We can find its modulus

$$\alpha = a + bi$$

$$|\alpha| = \sqrt{\alpha^* \alpha} = \sqrt{(a - bi)(a + bi)} = \sqrt{a^2 + b^2}$$

Example $\alpha = 3 + 2i$

:

$$|\alpha| = \sqrt{3^2 + 2^2} = \sqrt{13}$$

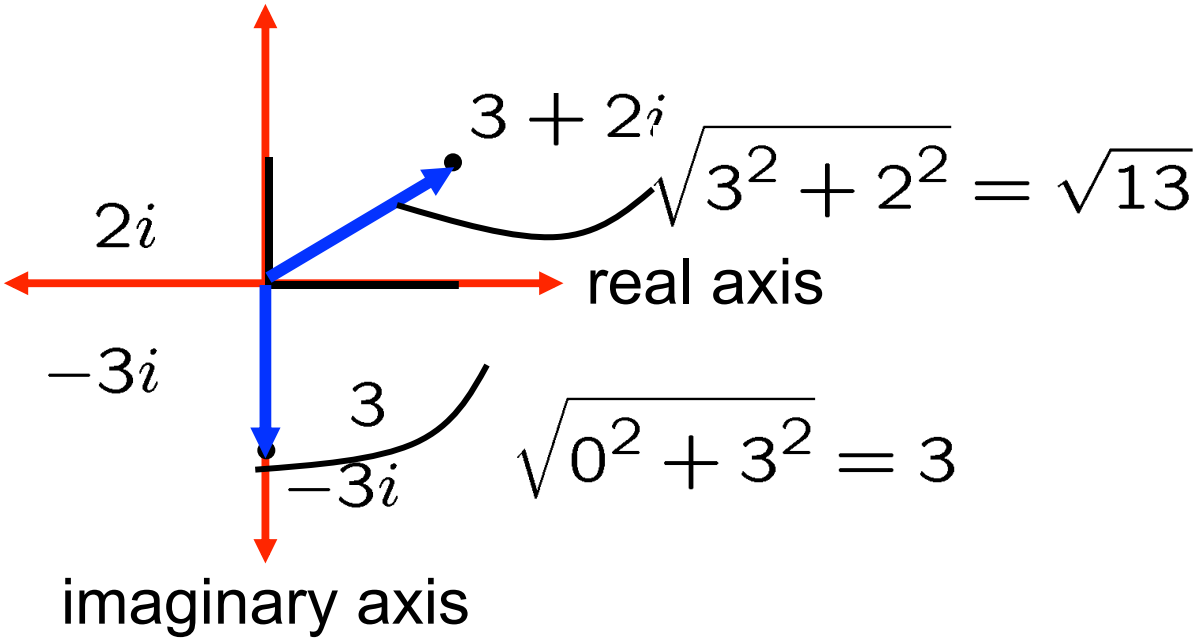
Complex Numbers, Modulus

Modulus

$$\alpha = a + bi$$

$$|\alpha| = \sqrt{\alpha^* \alpha} = \sqrt{(a - bi)(a + bi)} = \sqrt{a^2 + b^2}$$

Modulus is the length of the complex number in the complex plane:



Complex Numbers, Euler

Euler's formula

$$e^{i\theta} = \cos(\theta) + \sin(\theta)i$$

Example:

$$e^{\frac{i\pi}{4}} = \cos\left(\frac{\pi}{4}\right) + \sin\left(\frac{\pi}{4}\right)i = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i$$

The modulus of $e^{i\theta} = \cos(\theta) + \sin(\theta)i$

$$(e^{i\theta})^* = (\cos(\theta)+\sin(\theta)i)^* = \cos(\theta)-\sin(\theta)i = e^{-i\theta}$$

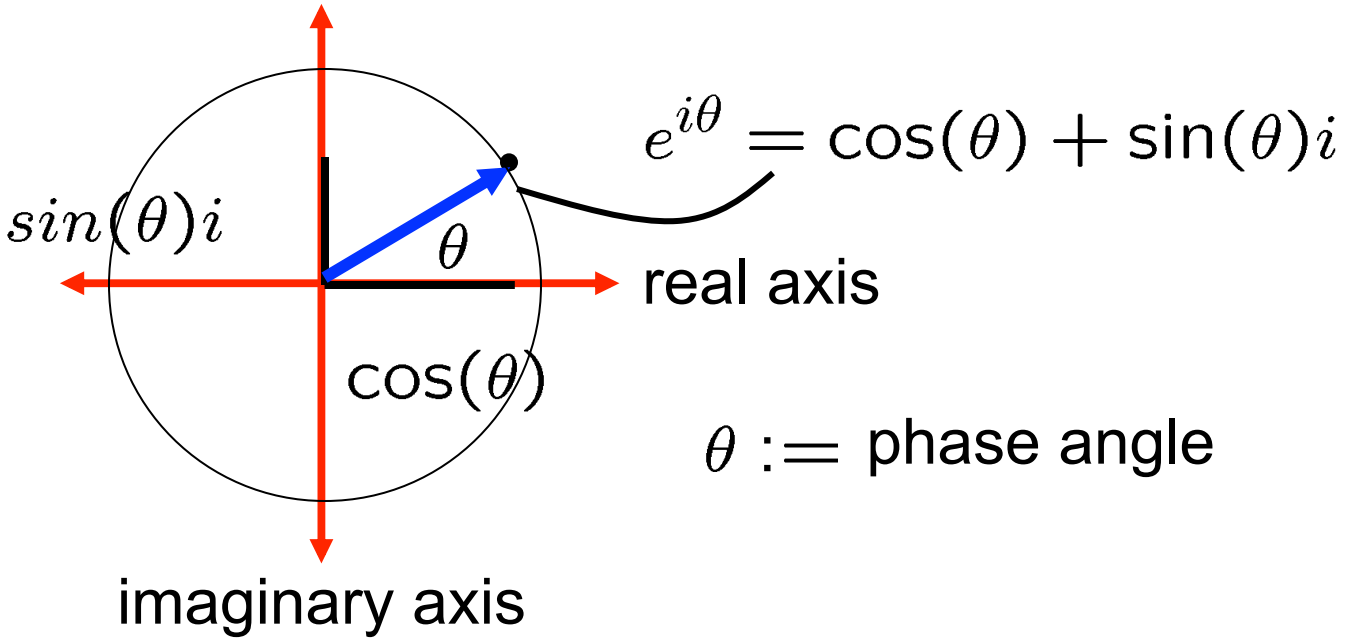
$$|e^{i\theta}| = \sqrt{(e^{i\theta})^* e^{i\theta}} = \cos^2\theta + \sin^2\theta = 1$$

Some important cases:

$$e^{\frac{i\pi}{2}} = i \qquad e^{i\pi} = -1 \qquad e^{i2\pi} = 1$$

Complex Numbers, Phases

Euler's formula geometrically



$\theta :=$ phase angle

Multiplying phases is beautiful:

$$e^{i\theta_1} e^{i\theta_2} = e^{i(\theta_1+\theta_2)}$$

Conjugating phases is also beautiful:

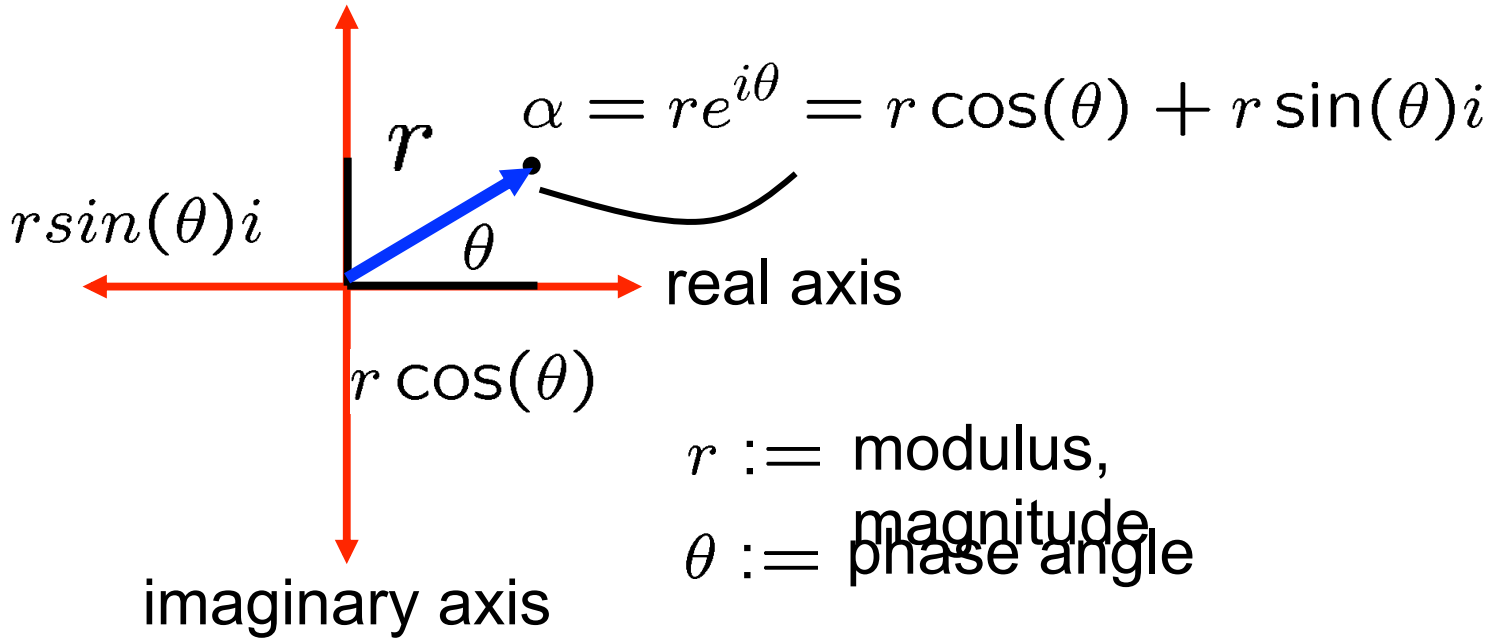
$$(e^{i\theta_1})^* = (\cos\theta + \sin\theta i)^* = \cos\theta - \sin\theta i = e^{-i\theta_1}$$

Complex Numbers, Geometry

All complex numbers can be expressed as:

$$\alpha = re^{i\theta} = r\cos(\theta) + r\sin(\theta)i$$

$$r = |\alpha| \qquad \tan\theta = \frac{Im(\alpha)}{Re(\alpha)}$$



$r :=$ modulus, magnitude

$\theta :=$ phase angle

Complex Numbers, Geometry

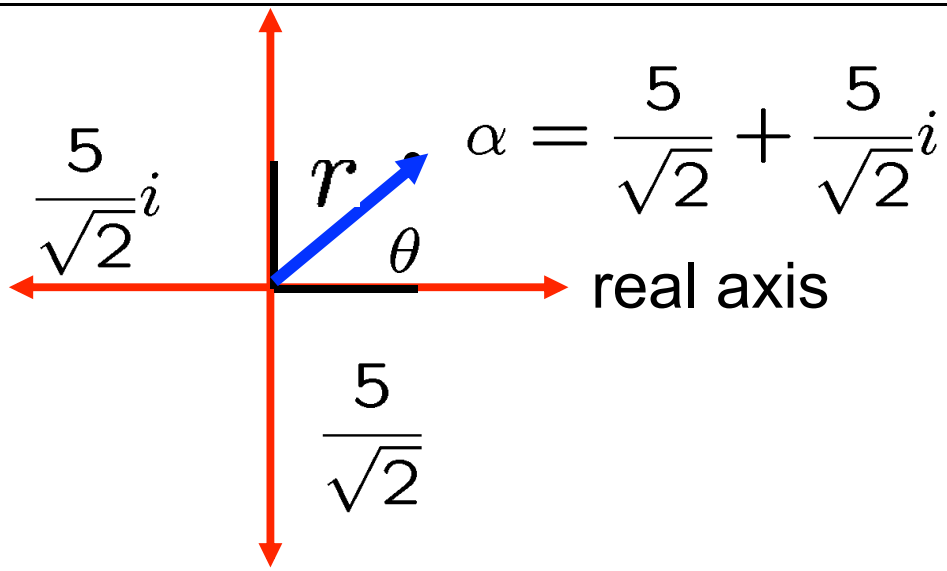
All complex numbers can be expressed as:

$$\alpha = re^{i\theta} = r\cos(\theta) + r\sin(\theta)i$$

Example:

$$\alpha = \frac{5}{\sqrt{2}} + \frac{5}{\sqrt{2}}i \qquad r = \sqrt{\alpha^*\alpha} = \sqrt{25/2 + 25/2} = 5$$

$$\alpha = 5e^{\frac{i\pi}{4}} \qquad \tan\theta = \frac{\frac{5}{\sqrt{2}}}{\frac{5}{\sqrt{2}}} = 1 \qquad \theta = \frac{\pi}{4}$$



$$\alpha = \frac{5}{\sqrt{2}} + \frac{5}{\sqrt{2}}i$$

Complex Numbers, Multiplying

All complex numbers can be expressed as:

$$\alpha = re^{i\theta} = r\cos(\theta) + r\sin(\theta)i$$

It is easy to multiply complex numbers when they are in this form

$$\alpha = r_1 e^{i\theta_1} \qquad \beta = r_2 e^{i\theta_2}$$

$$\alpha\beta = r_1 r_2 e^{i(\theta_1+\theta_2)}$$

Example
:

$$\alpha = 4e^{\frac{i\pi}{3}} \qquad \beta = 2e^{\frac{-i5\pi}{3}}$$

$$\alpha\beta = 4 \cdot 2e^{i\left(\frac{\pi}{3}+\frac{-5\pi}{3}\right)} = 8e^{\frac{-i4\pi}{3}}$$

Complex Vectors

N dimensional complex vector is a list of N complex numbers:

$$|v\rangle = \begin{bmatrix} v_0 \\ v_1 \\ \vdots \\ v_{N-1} \end{bmatrix}$$

v_i is the ith component of the vector

$$v_j = Re(v_j) + Im(v_j)i$$

"ket" "column vector"

(we start counting at 0 because eventually N will be a a power of 2)

Example:

3 dimensional complex vectors

$$|v\rangle = \begin{bmatrix} 1 \\ 2+3i \\ \sqrt{2}e^{\frac{i\pi}{5}} \end{bmatrix} \quad |v\rangle = \begin{bmatrix} 0 \\ -2i \\ 0 \end{bmatrix} \quad |v\rangle = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Complex Vectors, Scalar Times

Complex numbers can be multiplied by a complex number

$$|v\rangle = \begin{bmatrix} v_0 \\ v_1 \\ \vdots \\ v_{N-1} \end{bmatrix} \quad \alpha|v\rangle = \begin{bmatrix} \alpha v_0 \\ \alpha v_1 \\ \vdots \\ \alpha v_{N-1} \end{bmatrix}$$

α is a complex number

Example:

3 dimensional complex vector multiplied by a complex number

$$|v\rangle = \begin{bmatrix} 1 \\ 2+3i \\ \sqrt{2}e^{\frac{i\pi}{5}} \end{bmatrix} \quad (2i)|v\rangle = \begin{bmatrix} 2i \\ 2i(2+3i) \\ (2i)\sqrt{2}e^{\frac{i\pi}{5}} \end{bmatrix} = \begin{bmatrix} 2i \\ -5+4i \\ 2\sqrt{2}e^{\frac{i7\pi}{10}} \end{bmatrix}$$

Complex Vectors, Addition

Complex numbers can be added

$$|v\rangle = \begin{bmatrix} v_0 \\ v_1 \\ \vdots \\ v_{N-1} \end{bmatrix} \quad |w\rangle = \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_{N-1} \end{bmatrix}$$

$$|v\rangle + |w\rangle = \begin{bmatrix} v_0 + w_0 \\ v_1 + w_1 \\ \vdots \\ v_{N-1} + w_{N-1} \end{bmatrix}$$

Addition and multiplication by a scalar:

$$\alpha|v\rangle + \beta|w\rangle = \begin{bmatrix} \alpha v_0 + \beta w_0 \\ \alpha v_1 + \beta w_1 \\ \vdots \\ \alpha v_{N-1} + \beta w_{N-1} \end{bmatrix}$$

Complex Vectors, Addition

Examples
:

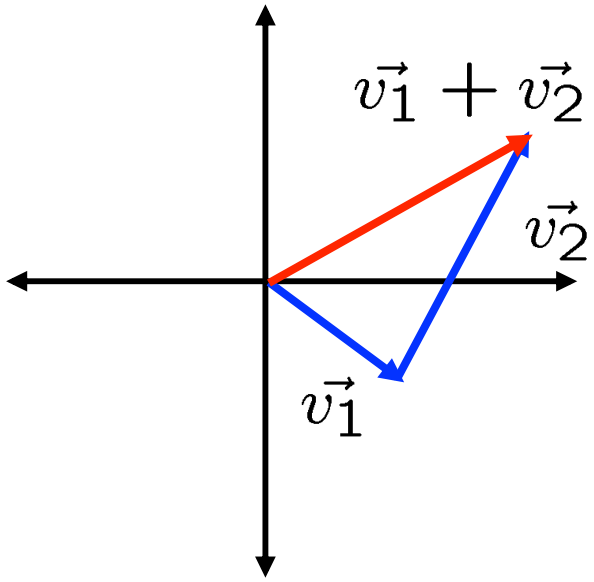
$$|v\rangle = \begin{bmatrix} 1 \\ 2+3i \\ \sqrt{2}e^{\frac{i\pi}{5}} \end{bmatrix} \quad |w\rangle = \begin{bmatrix} 4i \\ 2 \\ 0 \end{bmatrix}$$

$$|v\rangle + |w\rangle = \begin{bmatrix} 1+4i \\ 2+3i+2 \\ \sqrt{2}e^{\frac{i\pi}{5}} \end{bmatrix} = \begin{bmatrix} 1+4i \\ 4+3i \\ \sqrt{2}e^{\frac{i\pi}{5}} \end{bmatrix}$$

$$|v\rangle - i|w\rangle = \begin{bmatrix} 1-i4i \\ 2+3i-i2 \\ \sqrt{2}e^{\frac{i\pi}{5}} \end{bmatrix} = \begin{bmatrix} 5 \\ 2+i \\ \sqrt{2}e^{\frac{i\pi}{5}} \end{bmatrix}$$

Vectors, Addition

Remember adding real vectors looks geometrically like:



We should have a similar picture in mind for complex vectors

But the components of our vector are now complex numbers

Computational Basis

Some special vectors:

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix} \quad \cdots \quad |N-1\rangle = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}$$

Example:

2 dimensional complex vectors (also known as: a qubit!)

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \qquad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Computational Basis

Vectors can be “expanded” in the computational basis:

$$|v\rangle = \begin{bmatrix} v_0 \\ v_1 \\ \vdots \\ v_{N-1} \end{bmatrix} = v_0 \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} + v_1 \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix} + v_{N-1} \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}$$

$$= v_0|0\rangle + v_1|1\rangle + \cdots + v_{N-1}|N-1\rangle$$

Example:

$$|v\rangle = \begin{bmatrix} 1+2i \\ 3 \end{bmatrix} = (1+2i) \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 3 \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$= (1+2i)|0\rangle + 3|1\rangle$$

Computational Basis Math

$$|v\rangle = v_0|0\rangle + v_1|1\rangle + \cdots + v_{N-1}|N-1\rangle$$

$$|w\rangle = w_0|0\rangle + w_1|1\rangle + \cdots + w_{N-1}|N-1\rangle$$

$$|v\rangle + |w\rangle =$$

$$(v_0+w_0)|0\rangle+(v_1+w_1)|1\rangle+\cdots+(v_{N-1}+w_{N-1})|N-1\rangle$$

Example:

$$|v\rangle = |0\rangle + (2-3i)|1\rangle$$

$$|w\rangle = i|0\rangle + 4|1\rangle$$

$$|v\rangle+|w\rangle = (1+i)|0\rangle+(2-3i+4)|1\rangle = (1+i)|0\rangle+(6-3i)|1\rangle$$

Computational Basis Math

$$|v\rangle = v_0|0\rangle + v_1|1\rangle + \cdots + v_{N-1}|N-1\rangle$$

$$\alpha|v\rangle = \alpha v_0|0\rangle + \alpha v_1|1\rangle + \cdots + \alpha v_{N-1}|N-1\rangle$$

Example:

$$|v\rangle = |0\rangle + (2-3i)|1\rangle$$

$$|w\rangle = i|0\rangle + 4|1\rangle$$

$$i|w\rangle = i(i|0\rangle + 4|1\rangle) = (i^2)|0\rangle + (4i)|1\rangle = -|0\rangle + 4i|1\rangle$$

$$|v\rangle + i|w\rangle = (1-1)|0\rangle + (2-3i+4i)|1\rangle = (0)|0\rangle + (2+i)|1\rangle$$

$$= (2+i)|1\rangle$$

Bras and Kets

For every “ket,” there is a corresponding “bra” & vice versa

$$|v\rangle = \begin{bmatrix} v_0 \\ v_1 \\ \vdots \\ v_{N-1} \end{bmatrix} \qquad \langle v| = \begin{bmatrix} v_0^* & v_1^* & \cdots v_{N-1}^* \end{bmatrix}$$

Examples:

$$|v\rangle = \begin{bmatrix} 1 \\ 2+3i \\ \sqrt{2}e^{\frac{i\pi}{5}} \end{bmatrix} \qquad \langle v| = \begin{bmatrix} 1 & (2+3i)^* & (\sqrt{2}e^{\frac{i\pi}{5}})^* \end{bmatrix} = \begin{bmatrix} 1 & 2-3i & \sqrt{2}e^{\frac{-i\pi}{5}} \end{bmatrix}$$

$$\langle w| = \begin{bmatrix} 0 & 4i & 2+2i \end{bmatrix} \quad |w\rangle = \begin{bmatrix} 0 \\ (4i)^* \\ (2+2i)^* \end{bmatrix} = \begin{bmatrix} 0 \\ -4i \\ 2-2i \end{bmatrix}$$

Bras, Math

Multiplied by complex number

$$\langle v| = \begin{bmatrix} v_0^* & v_1^* & \cdots v_{N-1}^* \end{bmatrix} \quad \alpha\langle v| = \begin{bmatrix} \alpha v_0^* & \alpha v_1^* & \cdots \alpha v_{N-1}^* \end{bmatrix}$$

Example $\langle v| = \begin{bmatrix} 2 & 3+2i \end{bmatrix}$

$$(4+i)\langle v| = \begin{bmatrix} (4+i)2 & (4+i)(3+2i) \end{bmatrix} = \begin{bmatrix} 8+2i & 10+11i \end{bmatrix}$$

Added

$$\langle v| = \begin{bmatrix} v_0^* & v_1^* & \cdots v_{N-1}^* \end{bmatrix} \quad \langle w| = \begin{bmatrix} w_0^* & w_1^* & \cdots w_{N-1}^* \end{bmatrix}$$

$$\langle v| + \langle w| = \begin{bmatrix} v_0^* + w_0^* & v_1^* + w_1^* & \cdots v_{N-1}^* + w_{N-1}^* \end{bmatrix}$$

Example $\langle v| = \begin{bmatrix} 2 & 3+2i \end{bmatrix} \quad \langle w| = \begin{bmatrix} 1 & i \end{bmatrix}$

$$\langle v| + 2\langle w| = \begin{bmatrix} 2 & 3+2i \end{bmatrix} + \begin{bmatrix} 2 & 2i \end{bmatrix} = \begin{bmatrix} 4 & 2+4i \end{bmatrix}$$

Computational Bras

Computational Basis, but now for bras:

$$\langle 0| = \begin{bmatrix} 1 & 0 & \cdots 0 \end{bmatrix}$$

$$\langle 1| = \begin{bmatrix} 0 & 1 & \cdots 0 \end{bmatrix}$$

$$\vdots$$

$$\langle N-1| = \begin{bmatrix} 0 & 0 & \cdots 1 \end{bmatrix}$$

$$\langle v| = \begin{bmatrix} v_0^* & v_1^* & \cdots v_{N-1}^* \end{bmatrix} = v_0^*\langle 0| + v_1^*\langle 1| + \cdots + v_{N-1}^*\langle N-1|$$

Example:

$$\langle v| = \begin{bmatrix} 2 & 3+2i \end{bmatrix} = 2\langle 0| + (3+2i)\langle 1|$$

The Inner Product

Given a “bra” and a “ket” we can calculate an “inner product”

$$\langle w| = \begin{bmatrix} w_0^* & w_1^* & \cdots w_{N-1}^* \end{bmatrix} \qquad |v\rangle = \begin{bmatrix} v_0 \\ v_1 \\ \vdots \\ v_{N-1} \end{bmatrix}$$

$$\langle w|v\rangle = \begin{bmatrix} w_0^* & w_1^* & \cdots w_{N-1}^* \end{bmatrix} \begin{bmatrix} v_0 \\ v_1 \\ \vdots \\ v_{N-1} \end{bmatrix}$$

$$= w_0^* v_0 + w_1^* v_1 + \cdots + w_{N-1}^* v_{N-1}$$

This is a generalization of the dot product for real vectors

The result of taking an inner product is a complex number

The Inner Product

$$\langle w|v\rangle = w_0^* v_0 + w_1^* v_1 + \cdots + w_{N-1}^* v_{N-1}$$

Example:

$$|v\rangle = \begin{bmatrix} 1 \\ 1+2i \end{bmatrix} \qquad |w\rangle = \begin{bmatrix} 3i \\ 3 \end{bmatrix}$$

$$\langle w| = \begin{bmatrix} -3i & 3 \end{bmatrix}$$

$$\langle w|v\rangle = \begin{bmatrix} -3i & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 1+2i \end{bmatrix} = (-3i)\cdot 1 + 3(1+2i) = 3+3i$$

$$\langle v|w\rangle = \begin{bmatrix} 1 & 1-2i \end{bmatrix} \begin{bmatrix} 3i \\ 3 \end{bmatrix} = 1\cdot(3i) + (1-2i)3 = 3-3i$$

Complex conjugate of inner product: $(\langle w|v\rangle)^* = \langle v|w\rangle$

The Inner Product in Comp. Basis

$$\langle w|v\rangle = w_0^* v_0 + w_1^* v_1 + \cdots + w_{N-1}^* v_{N-1}$$

$$\langle 0|0\rangle = \begin{bmatrix} 1 & 0 & \cdots 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = 1{\cdot}1 + 0{\cdot}0 + \cdots + 0{\cdot}0 = 1$$

$$\langle 0|1\rangle = \begin{bmatrix} 1 & 0 & \cdots 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix} = 1{\cdot}0 + 1{\cdot}0 + \cdots + 0{\cdot}0 = 0$$

Inner product of computational basis elements:

$$\langle j|k\rangle = \delta_{j,k} = \begin{cases} 1 \text{ if } j = k \\ 0 \text{ otherwise} \end{cases}$$

Kronecker delta

The Inner Product in Comp. Basis

$$\langle w|v\rangle = w_0^* v_0 + w_1^* v_1 + \cdots + w_{N-1}^* v_{N-1}$$

$$\langle w| = w_0^*\langle 0| + w_1^*\langle 1| + \cdots + w_{N-1}^*\langle N-1|$$

$$|v\rangle = v_0|0\rangle + v_1|1\rangle + \cdots + v_{N-1}|N-1\rangle$$

$$\begin{aligned}\langle w|v\rangle \;=\; & (w_0^*\langle 0| + w_1^*\langle 1| + \cdots + w_{N-1}^*\langle N-1|) \\ & (v_0|0\rangle + v_1|1\rangle + \cdots + v_{N-1}|N-1\rangle)\end{aligned}$$

$$\langle w|v\rangle = w_0^* v_0 + w_1^* v_1 + \cdots + w_{N-1}^* v_{N-1}$$

Example $|v\rangle = |0\rangle + 2i|1\rangle$ $|w\rangle = 3i|0\rangle + (2i+2)|1\rangle$

:

$$\langle w|v\rangle = -3i \cdot 1 + (-2i+2)2i = 4 + i$$

In Class Problem # 1

Norm of a Vector

Norm of a vector

$$|||v\rangle|| = \sqrt{\langle v|v\rangle}$$

$$\langle v|v\rangle = v_0^* v_0 + v_1^* v_1 + \cdots + v_{N-1}^* v_{N-1}$$
$$= |v_0|^2 + |v_1|^2 + \cdots + |v_{N-1}|^2$$

which is always a positive real number

it is (roughly) the length of the complex vector

Example $|v\rangle = |0\rangle + 2i|1\rangle$

:

$$\langle v|v\rangle = |1|^2 + |2i|^2 = 5$$

$$|||v\rangle|| = \sqrt{5}$$

Quantum Rule 1

Rule 1: The wave function of a N dimensional quantum system is given by an N dimensional complex vector with norm equal to one.

$$|v\rangle = v_0|0\rangle + v_1|1\rangle + \cdots + v_{N-1}|N-1\rangle$$

$$|||v\rangle|| = 1$$

Example:

a valid wave function for a 3 dimensional quantum system

$$|v\rangle = \frac{1}{\sqrt{6}}|0\rangle + \frac{1}{\sqrt{3}}|1\rangle + \frac{i}{\sqrt{2}}|2\rangle$$

$$|||v\rangle|| = \sqrt{\left|\frac{1}{\sqrt{6}}\right|^2 + \left|\frac{1}{\sqrt{3}}\right|^2 + \left|\frac{i}{\sqrt{2}}\right|^2} = \sqrt{\frac{1}{6} + \frac{1}{3} + \frac{1}{2}} = 1$$

Qubits

Two dimensional quantum systems are called qubits

A qubit has a wave function which we write as

$$|v\rangle = v_0|0\rangle + v_1|1\rangle \qquad |v_0|^2 + |v_1|^2 = 1$$

Examples:

Valid qubit wave functions:

$$|v\rangle = |0\rangle \qquad |||v\rangle|| = \sqrt{|1|^2 + |0|^2} = 1$$

$$|v\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{i}{\sqrt{2}}|1\rangle \quad |||v\rangle|| = \sqrt{\left|\frac{1}{\sqrt{2}}\right|^2 + \left|\frac{i}{\sqrt{2}}\right|^2} = 1$$

Invalid qubit wave function:

$$|v\rangle = 5|0\rangle + i|1\rangle \qquad |||v\rangle|| = \sqrt{|5|^2 + |i|^2} = \sqrt{26}$$

Measuring Qubits

A bit is a classical system with two possible states, 0 and 1

A qubit is a quantum system with two possible states, 0 and 1

When we observe a qubit, we get the result 0 or the result 1

0	or	1

If before we observe the qubit the wave function of the qubit is

$$|v\rangle = v_0|0\rangle + v_1|1\rangle$$

then the probability that we observe 0 is $|v_0|^2$

and the probability that we observe 1 is $|v_1|^2$

"measuring in the computational basis"

Measuring Qubits

Example:

We are given a qubit with wave function

$$|v\rangle = \frac{1}{\sqrt{3}}|0\rangle + i\sqrt{\frac{2}{3}}|1\rangle \quad |||v\rangle|| = \sqrt{\left|\frac{1}{\sqrt{3}}\right|^2 + \left|i\sqrt{\frac{2}{3}}\right|^2} = 1$$

If we observe the system in the computational basis, then we get outcome 0 with probability

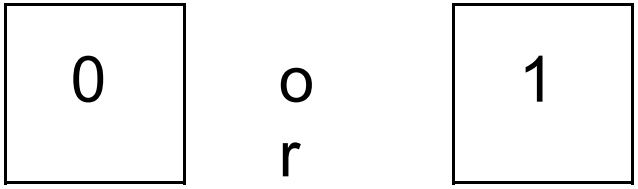
$$\left|\frac{1}{\sqrt{3}}\right|^2 = \frac{1}{3}$$

and we get outcome 1 with probability:

$$\left|\frac{\sqrt{2}i}{\sqrt{3}}\right|^2 = \frac{2}{3}$$

Measuring Qubits Continued

When we observe a qubit, we get the result 0 or the result 1



If before we observe the qubit the wave function of the qubit is

$$|v\rangle = v_0|0\rangle + v_1|1\rangle$$

then the probability that we observe 0 is $|v_0|^2$

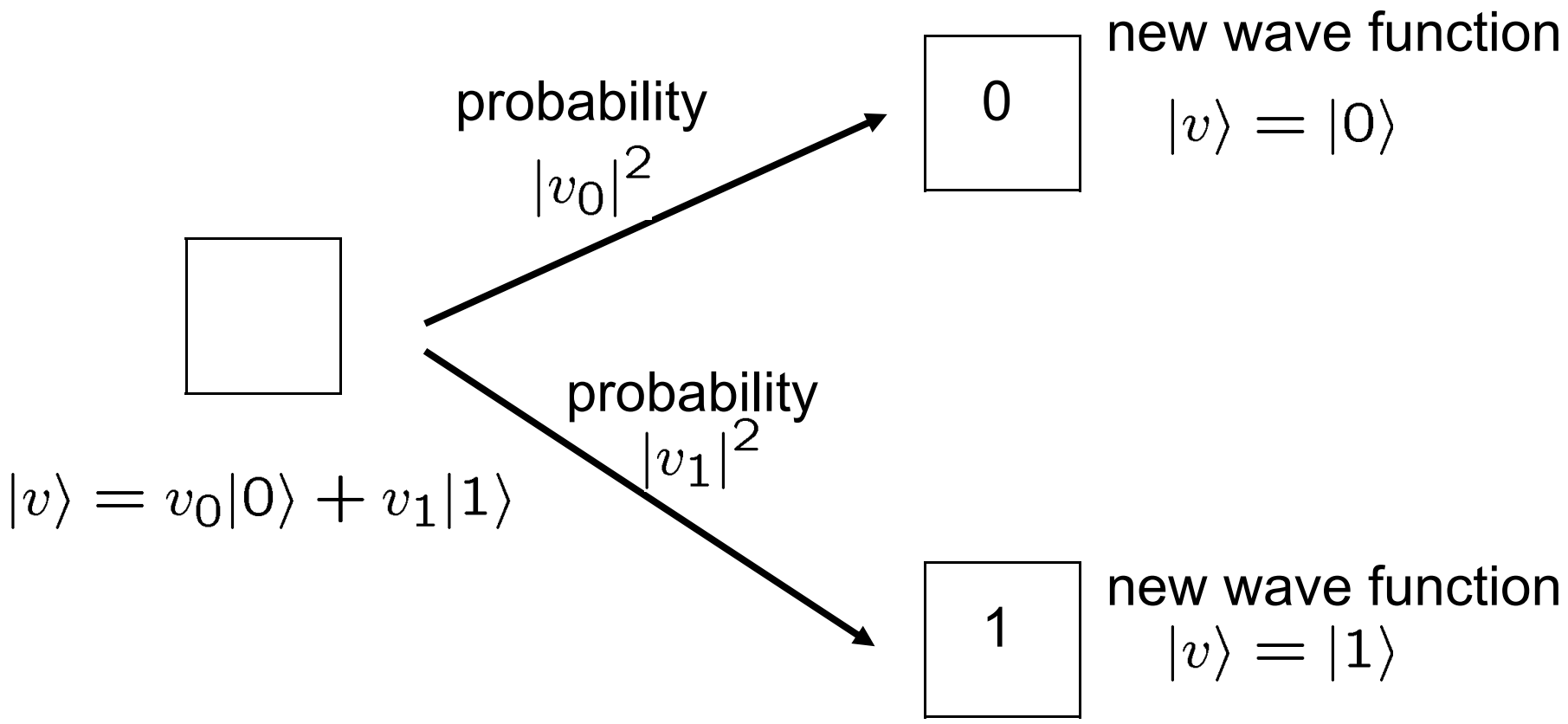
and the new wave function for the qubit is $|0\rangle$

and the probability that we observe 1 is $|v_1|^2$

and the new wave function for the qubit is $|1\rangle$

"measuring in the computational basis"

Measuring Qubits Continued



The wave function is a description of our system.

When we measure the system we find the system in one state

This happens with probabilities we get from our description

Measuring Qubits

Example:

We are given a qubit with wave function

$$|v\rangle = \frac{1}{\sqrt{3}}|0\rangle + i\sqrt{\frac{2}{3}}|1\rangle \quad |||v\rangle|| = \sqrt{\left|\frac{1}{\sqrt{3}}\right|^2 + \left|i\sqrt{\frac{2}{3}}\right|^2} = 1$$

If we observe the system in the computational basis, then we get outcome 0 with probability

$$\left|\frac{1}{\sqrt{3}}\right|^2 = \frac{1}{3}$$ new wave function $|0\rangle$

and we get outcome 1 with probability:

$$\left|\frac{2i}{\sqrt{3}}\right|^2 = \frac{2}{3}$$ new wave function $|1\rangle$

Measuring Qubits

Example:

We are given a qubit with wave function

$$|v\rangle = |0\rangle \qquad |||v\rangle|| = \sqrt{|1|^2 + |0|^2} = 1$$

If we observe the system in the computational basis, then we get outcome 0 with probability

$$|1|^2 = 1$$ new wave function $|0\rangle$

and we get outcome 1 with probability:

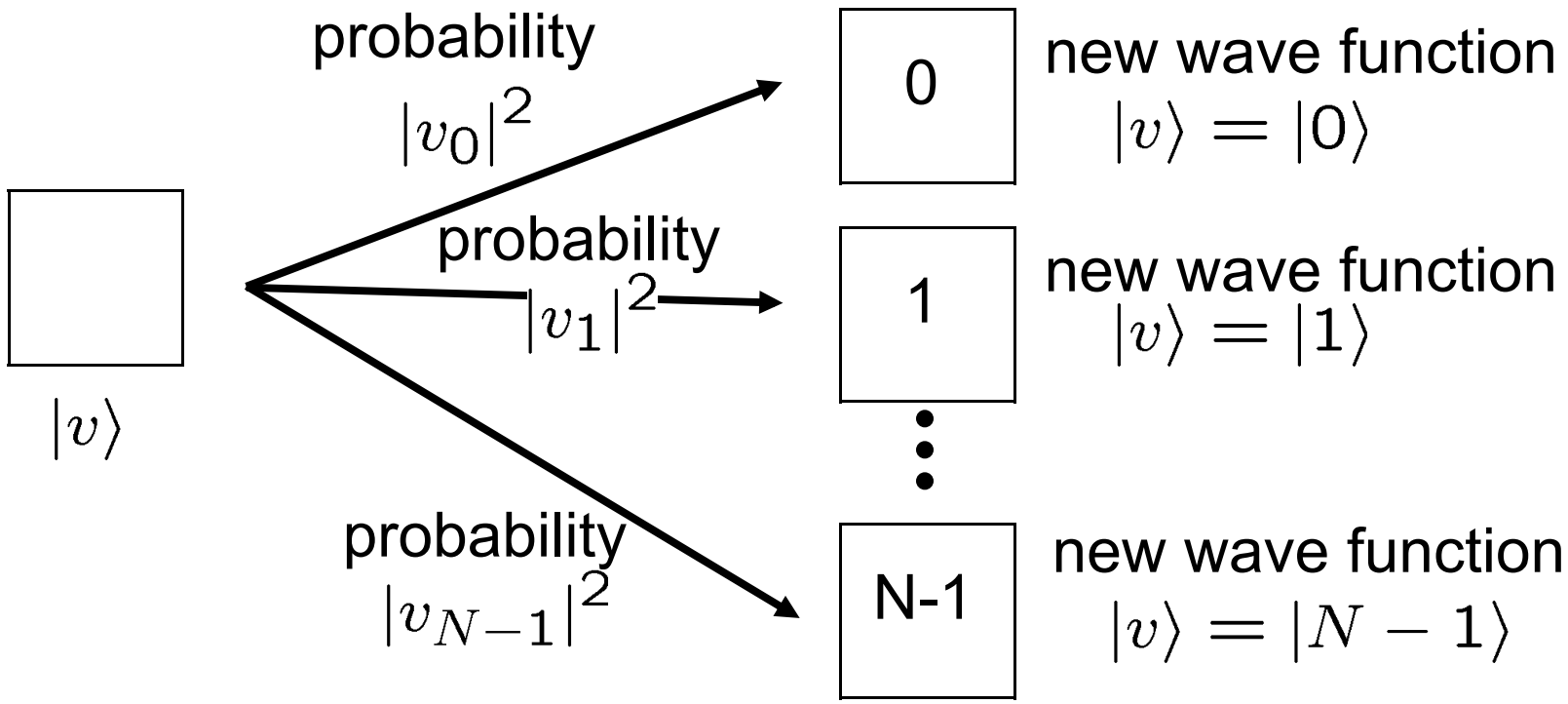
$$|0|^2 = 0$$ a.k.a never

Quantum Rule 3

Rule 3: If we measure a N dimensional quantum system with the wave function

$$|v\rangle = v_0|0\rangle + v_1|1\rangle + \cdots + v_{N-1}|N-1\rangle$$

in the $|0\rangle, |1\rangle, \ldots, |N-1\rangle$ basis, then the probability of observing the system in the state $|j\rangle$ is $|v_j|^2$. After such a measurement, the wave function of the system is $|j\rangle$



Matrices

A N dimensional complex matrix M is an N by N array of complex numbers:

$$M = \begin{bmatrix} M_{0,0} & \cdots & M_{0,N-1} \\ \vdots & & \vdots \\ M_{N-1,0} & \cdots & M_{N-1,N-1} \end{bmatrix}$$

$M_{j,k}$ are complex numbers

Example

- Three dimensional complex matrix:

$$M = \begin{bmatrix} 4 & 3+i & 2 \\ i & e^{\frac{\pi}{4}} & \sqrt{2}i \\ 0 & 0 & 4 \end{bmatrix}$$

$$M_{1,0} = i$$

$$M_{2,2} = 4$$

Matrices, Multiplied by Scalar

Matrices can be multiplied by a complex number

$$M = \begin{bmatrix} M_{0,0} & \cdots & M_{0,N-1} \\ \vdots & & \vdots \\ M_{N-1,0} & \cdots & M_{N-1,N-1} \end{bmatrix}$$

$$\alpha M = \begin{bmatrix} \alpha M_{0,0} & \cdots & \alpha M_{0,N-1} \\ \vdots & & \vdots \\ \alpha M_{N-1,0} & \cdots & \alpha M_{N-1,N-1} \end{bmatrix}$$

Example:

$$M = \begin{bmatrix} 0 & 3+i \\ i & 1 \end{bmatrix} \qquad \alpha = 2i$$

$$\alpha M = \begin{bmatrix} 2i \cdot 0 & 2i(3+i) \\ 2i(i) & 2i(1) \end{bmatrix} = \begin{bmatrix} 0 & -2+6i \\ -2 & 2i \end{bmatrix}$$

Matrices, Added

Matrices can be added

$$M = \begin{bmatrix} M_{0,0} & \cdots & M_{0,N-1} \\ \vdots & & \vdots \\ M_{N-1,0} & \cdots & M_{N-1,N-1} \end{bmatrix} L = \begin{bmatrix} L_{0,0} & \cdots & L_{0,N-1} \\ \vdots & & \vdots \\ L_{N-1,0} & \cdots & L_{N-1,N-1} \end{bmatrix}$$

$$M+L = \begin{bmatrix} M_{0,0}+L_{0,0} & \cdots & M_{0,N-1}+L_{0,N-1} \\ \vdots & & \vdots \\ M_{N-1,0}+L_{N-1,0} & \cdots & M_{N-1,N-1}+L_{N-1,N-1} \end{bmatrix}$$

Example:

$$M = \begin{bmatrix} 0 & 3+i \\ i & 1 \end{bmatrix} \qquad L = \begin{bmatrix} 3 & -3-i \\ i & 2 \end{bmatrix}$$

$$M+L = \begin{bmatrix} 0+3 & 3+i+(-3-i) \\ i+i & 1+2 \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 2i & 3 \end{bmatrix}$$

Matrices, Multiplied

Matrices can be multiplied

$$M = \begin{bmatrix} M_{0,0} & \cdots & M_{0,N-1} \\ \vdots & & \vdots \\ M_{N-1,0} & \cdots & M_{N-1,N-1} \end{bmatrix} L = \begin{bmatrix} L_{0,0} & \cdots & L_{0,N-1} \\ \vdots & & \vdots \\ L_{N-1,0} & \cdots & L_{N-1,N-1} \end{bmatrix}$$

$$R = ML = \begin{bmatrix} R_{0,0} & \cdots & R_{0,N-1} \\ \vdots & & \vdots \\ R_{N-1,0} & \cdots & R_{N-1,N-1} \end{bmatrix}$$

$$R_{0,0} = M_{0,0}L_{0,0} + M_{0,1}L_{1,0} + \cdots + M_{0,N-1}L_{N-1,0}$$
$$R_{0,1} = M_{0,0}L_{0,1} + M_{0,1}L_{1,1} + \cdots + M_{0,N-1}L_{N-1,1}$$
$$\vdots$$
$$R_{j,k} = M_{j,0}L_{0,k} + M_{j,1}L_{1,k} + \cdots + M_{j,N-1}L_{N-1,k}$$

$$R_{j,k} = \sum_{l=0}^{N-1} M_{j,l}L_{l,k}$$

Matrices, Multiplied

Example:

$$M = \begin{bmatrix} 0 & 3+i \\ i & 1 \end{bmatrix} \qquad L = \begin{bmatrix} 3 & -3-i \\ i & 2 \end{bmatrix}$$

$$ML = \begin{bmatrix} 0 \cdot 3 + (3+i)i & 0 \cdot (-3-i) + (3+i)2 \\ i(3) + 1(i) & i(-3-i) + 1(2) \end{bmatrix}$$

$$ML = \begin{bmatrix} -1+3i & 6+2i \\ 4i & 3-3i \end{bmatrix}$$

$$LM = \begin{bmatrix} 3 \cdot 0 + (-3-i)i & 3(3+i) + (-3-i)1 \\ i \cdot 0 + 2(i) & i(3+i) + 2(1) \end{bmatrix}$$

$$LM = \begin{bmatrix} 1-3i & 6+2i \\ 2i & 1+3i \end{bmatrix}$$

Note $ML \neq LM$

Matrices and Kets, Multiplied

Given a matrix, and a column vector:

$$M = \begin{bmatrix} M_{0,0} & \cdots & M_{0,N-1} \\ \vdots & & \vdots \\ M_{N-1,0} & \cdots & M_{N-1,N-1} \end{bmatrix} \qquad |v\rangle = \begin{bmatrix} v_0 \\ v_1 \\ \vdots \\ v_{N-1} \end{bmatrix}$$

These can be multiplied to obtain a new column vector:

$$M|v\rangle = \begin{bmatrix} M_{0,0} & \cdots & M_{0,N-1} \\ \vdots & & \vdots \\ M_{N-1,0} & \cdots & M_{N-1,N-1} \end{bmatrix} \begin{bmatrix} v_0 \\ \vdots \\ v_{N-1} \end{bmatrix}$$

$$= \begin{bmatrix} M_{0,0}v_0 + M_{0,1}v_1 + \cdots M_{0,N-1}v_{N-1} \\ \vdots \\ M_{N-1,0}v_0 + M_{N-1,1}v_1 + \cdots M_{N-1,N-1}v_{N-1} \end{bmatrix}$$

Matrices and Kets, Multiplied

Example:

$$M = \begin{bmatrix} 0 & 3+i \\ i & 1 \end{bmatrix} \quad |v\rangle = \begin{bmatrix} 3 \\ -2i \end{bmatrix} = 3|0\rangle - 2i|1\rangle$$

$$M|v\rangle = \begin{bmatrix} 0 & 3+i \\ i & 1 \end{bmatrix} \begin{bmatrix} 3 \\ -2i \end{bmatrix}$$

$$= \begin{bmatrix} 0 \cdot 3 + (3+i)(-2i) \\ i(3) + 1(-2i) \end{bmatrix} = \begin{bmatrix} 2-6i \\ i \end{bmatrix}$$

$$M|v\rangle = (2-6i)|0\rangle + i|1\rangle$$

Matrices and Bras, Multiplied

Given a matrix, and a row vector:

$$M = \begin{bmatrix} M_{0,0} & \cdots & M_{0,N-1} \\ \vdots & & \vdots \\ M_{N-1,0} & \cdots & M_{N-1,N-1} \end{bmatrix} \quad \langle w| = \begin{bmatrix} w_0^* & w_1^* & \cdots w_{N-1}^* \end{bmatrix}$$

These can be multiplied to obtain a new row vector:

$$\langle w|M = \begin{bmatrix} w_0^* & \cdots & w_{N-1}^* \end{bmatrix} \begin{bmatrix} M_{0,0} & \cdots & M_{0,N-1} \\ \vdots & & \vdots \\ M_{N-1,0} & \cdots & M_{N-1,N-1} \end{bmatrix}$$

$$\langle w|M = \begin{bmatrix} r_0^* & \cdots & r_{N-1}^* \end{bmatrix}$$

$$\begin{aligned} r_0^* &= w_0^* M_{0,0} + \cdots + w_{N-1}^* M_{N-1,0} \\ &\vdots \\ r_{N-1}^* &= w_0^* M_{0,N-1} + \cdots + w_{N-1}^* M_{N-1,N-1} \end{aligned}$$

Matrices and Bras, Multiplied

Example:

$$M = \begin{bmatrix} 0 & 3+i \\ i & 1 \end{bmatrix} \quad |w\rangle = \begin{bmatrix} 3 \\ -2i \end{bmatrix} = 3|0\rangle - 2i|1\rangle$$

$$\langle w| = \begin{bmatrix} 3 & 2i \end{bmatrix} = 3\langle 0| + 2i\langle 1|$$

$$\langle w|M = \begin{bmatrix} 3 & 2i \end{bmatrix} \begin{bmatrix} 0 & 3+i \\ i & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 3\cdot 0 + 2i(i) & 3(3+i) + 2i(1) \end{bmatrix}$$

$$= \begin{bmatrix} -2 & 9+5i \end{bmatrix} = -2\langle 0| + (9+6i)\langle 1|$$

Matrices, Complex Conjugate

Given a matrix, we can form its complex conjugate by conjugating every element:

$$M = \begin{bmatrix} M_{0,0} & \cdots & M_{0,N-1} \\ \vdots & & \vdots \\ M_{N-1,0} & \cdots & M_{N-1,N-1} \end{bmatrix}$$

$$M^* = \begin{bmatrix} M^*_{0,0} & \cdots & M^*_{0,N-1} \\ \vdots & & \vdots \\ M^*_{N-1,0} & \cdots & M^*_{N-1,N-1} \end{bmatrix}$$

Example:

$$M = \begin{bmatrix} 0 & 3+i \\ i & 1 \end{bmatrix} \qquad M^* = \begin{bmatrix} 0 & 3-i \\ -i & 1 \end{bmatrix}$$

Matrices, Transpose

Given a matrix, we can form it's transpose by reflecting across the diagonal

$$M = \begin{bmatrix} M_{0,0} & \cdots & M_{0,N-1} \\ \vdots & & \vdots \\ M_{N-1,0} & \cdots & M_{N-1,N-1} \end{bmatrix}$$

$$M^T = \begin{bmatrix} M_{0,0} & \cdots & M_{N-1,0} \\ \vdots & & \vdots \\ M_{0,N-1} & \cdots & M_{N-1,N-1} \end{bmatrix}$$

Example:

$$M = \begin{bmatrix} 0 & 3+i \\ i & 1 \end{bmatrix} \qquad M^T = \begin{bmatrix} 0 & i \\ 3+i & 1 \end{bmatrix}$$

Matrices, Conjugate Transpose

Given a matrix, we can form its conjugate transpose by reflecting across the diagonal and conjugating

$$M = \begin{bmatrix} M_{0,0} & \cdots & M_{0,N-1} \\ \vdots & & \vdots \\ M_{N-1,0} & \cdots & M_{N-1,N-1} \end{bmatrix}$$

$$M^\dagger = \begin{bmatrix} M^*_{0,0} & \cdots & M^*_{N-1,0} \\ \vdots & & \vdots \\ M^*_{0,N-1} & \cdots & M^*_{N-1,N-1} \end{bmatrix}$$

Example:

$$M = \begin{bmatrix} 0 & 3+i \\ i & 1 \end{bmatrix} \qquad M^\dagger = \begin{bmatrix} 0 & -i \\ 3-i & 1 \end{bmatrix}$$

Bras, Kets, Conjugate Transpose

Taking the conjugate transpose of a ket

$$|v\rangle = \begin{bmatrix} v_0 \\ v_1 \\ \vdots \\ v_{N-1} \end{bmatrix} \qquad (|v\rangle)^T = \begin{bmatrix} v_0 & v_1 & \cdots & v_{N-1} \end{bmatrix}$$

gives the corresponding bra:

$$(|v\rangle)^\dagger = \begin{bmatrix} v_0^* & v_1^* & \cdots & v_{N-1}^* \end{bmatrix} = \langle v|$$

Similarly we can take the conjugate transpose of a bra to get the corresponding ket:

$$(\langle v|)^\dagger = |v\rangle$$

Unitary Matrices

A matrix U is unitary if

$$U^\dagger U = I$$

N x N identity matrix

$$I = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & & \vdots \\ \vdots & & \ddots & \\ 0 & \cdots & & 1 \end{bmatrix} = diag(1, 1, \ldots, 1)$$

Equivalently a matrix U is unitary if

$$UU^\dagger = I$$

Unitary Example

$$U = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{-i}{\sqrt{2}} & \frac{i}{\sqrt{2}} \end{bmatrix}$$

Conjugate:

$$U^* = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{i}{\sqrt{2}} & \frac{-i}{\sqrt{2}} \end{bmatrix}$$

Conjugate transpose:

$$U^\dagger = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{-i}{\sqrt{2}} \end{bmatrix}$$

Unitary?

$$UU^\dagger = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{i}{\sqrt{2}} & \frac{i}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{-i}{\sqrt{2}} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{\sqrt{2}}\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\frac{i}{\sqrt{2}} + \frac{1}{\sqrt{2}}\frac{-i}{\sqrt{2}} \\ \frac{-i}{\sqrt{2}}\frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}}\frac{1}{\sqrt{2}} & \frac{-i}{\sqrt{2}}\frac{i}{\sqrt{2}} + \frac{i}{\sqrt{2}}\frac{-i}{\sqrt{2}} \end{bmatrix}$$

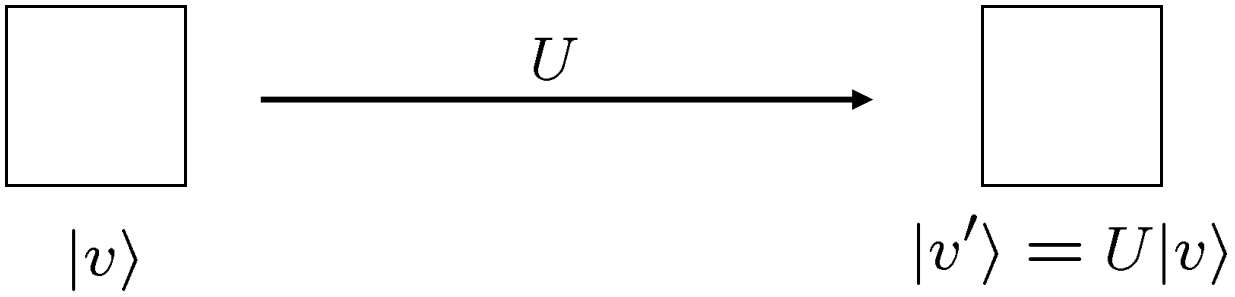
$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Yes: $UU^\dagger = I$

Quantum Rule 2

Rule 2: The wave function of a N dimensional quantum system evolves in time according to a unitary matrix U. If the wave function initially is $|v\rangle$ then after the evolution correspond to U the new wave function is $|v'\rangle = U|v\rangle$

"Unitary Evolution"



Unitary Evolution and the Norm

Unitary evolution

$$|v'\rangle = U|v\rangle$$

What happens to the norm $\langle v'|v'\rangle$ of the ket?

$$\langle v'| = (U|v\rangle)^\dagger = \langle v|U^\dagger$$

$$\langle v'|v'\rangle = \langle v|U^\dagger U|v\rangle = \langle v|I|v\rangle = \langle v|v\rangle$$

Unitary evolution does not change the length of the ket.

Normalized wave function

$$\sqrt{\langle v|v\rangle} = 1$$

$$|v'\rangle \xrightarrow{} U|v\rangle$$

unitary evolution

Normalized wave function

$$\sqrt{\langle v'|v'\rangle} = 1$$

Unitary Evolution for Qubits

Unitary evolution will be described by a two dimensional unitary matrix

$$U = \begin{bmatrix} U_{0,0} & U_{0,1} \\ U_{1,0} & U_{1,1} \end{bmatrix}$$

If initial qubit wave function is

$$|v\rangle = \begin{bmatrix} v_0 \\ v_1 \end{bmatrix} = v_0|0\rangle + v_1|1\rangle$$

Then this evolves to

$$|v'\rangle = \begin{bmatrix} U_{0,0} & U_{0,1} \\ U_{1,0} & U_{1,1} \end{bmatrix} \begin{bmatrix} v_0 \\ v_1 \end{bmatrix} = \begin{bmatrix} U_{0,0}v_0 + U_{0,1}v_1 \\ U_{1,0}v_0 + U_{1,1}v_1 \end{bmatrix}$$

Unitary Evolution for Qubits

$$U = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{-i}{\sqrt{2}} & \frac{i}{\sqrt{2}} \end{bmatrix} \qquad |v\rangle = \begin{bmatrix} \frac{1}{2} \\ \frac{\sqrt{3}}{2} \end{bmatrix} = \frac{1}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle$$

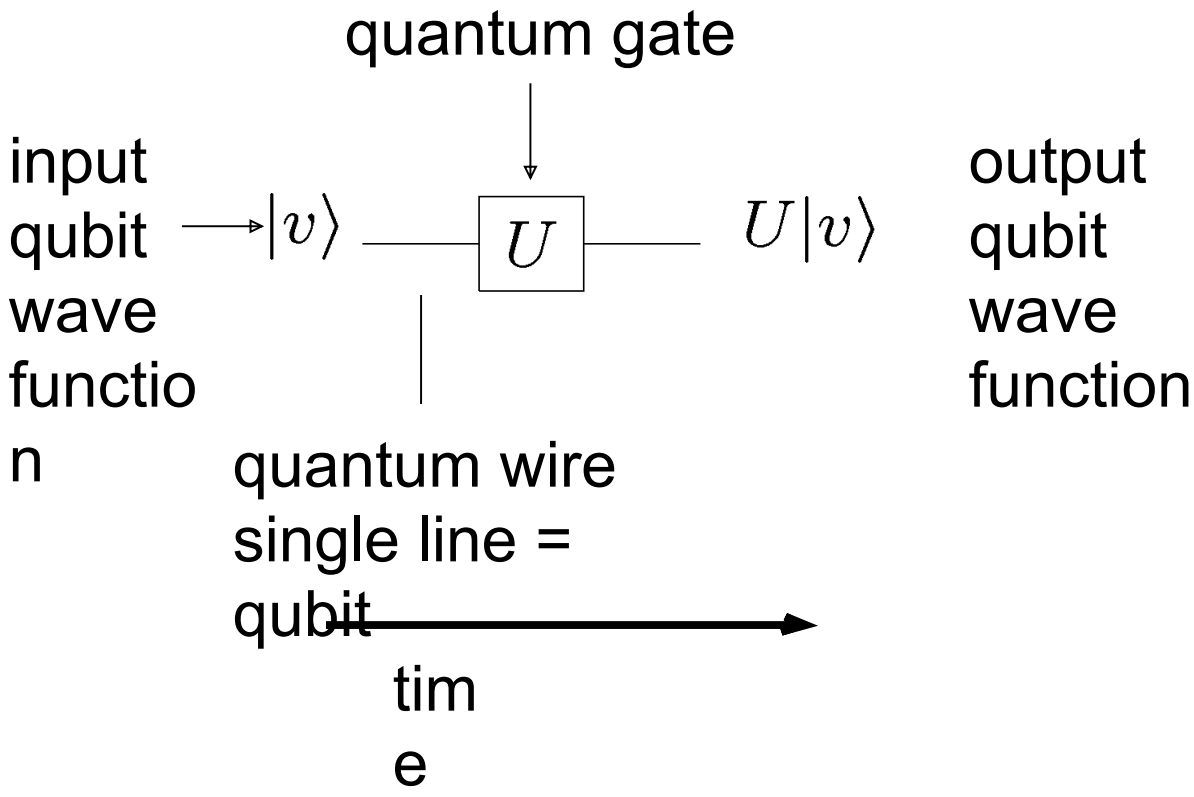
$$|v'\rangle = U|v\rangle = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{-i}{\sqrt{2}} & \frac{i}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{1}{2} \\ \frac{\sqrt{3}}{2} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{\sqrt{2}}\frac{1}{2} + \frac{1}{\sqrt{2}}\frac{\sqrt{3}}{2} \\ \frac{-i}{\sqrt{2}}\frac{1}{2} + \frac{i}{\sqrt{2}}\frac{\sqrt{3}}{2} \end{bmatrix} = \begin{bmatrix} \frac{1+\sqrt{3}}{2\sqrt{2}} \\ \frac{(-1+\sqrt{3})i}{2\sqrt{2}} \end{bmatrix}$$

$$|v'\rangle = \frac{1+\sqrt{3}}{2\sqrt{2}}|0\rangle + \frac{(-1+\sqrt{3})i}{2\sqrt{2}}|1\rangle$$

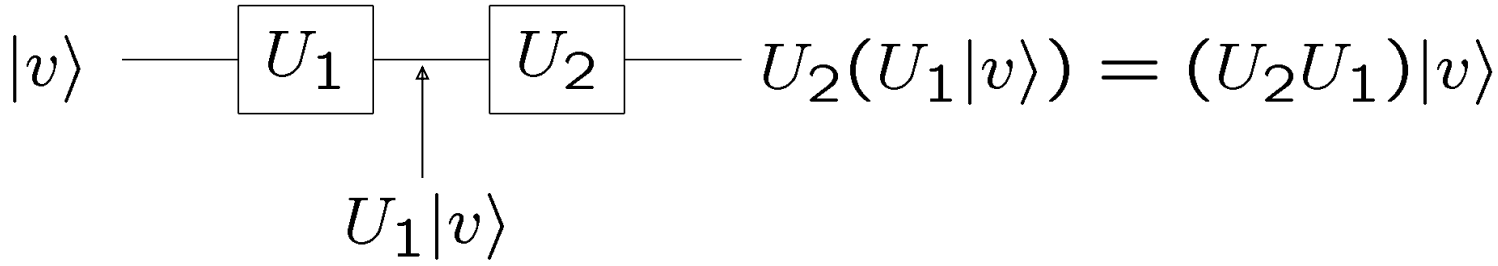
Single Qubit Quantum Circuits

Circuit diagrams for evolving qubits

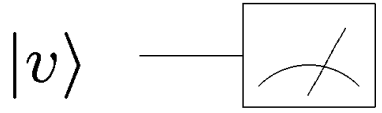


Single Qubit Quantum Circuits

Two unitary evolutions:



measurement in the $|0\rangle, |1\rangle$ basis



$|v\rangle = v_0|0\rangle + v_1|1\rangle$

Probability of outcome 0: $|v_0|^2$

Probability of outcome 1: $|v_1|^2$

In Class Problem #2